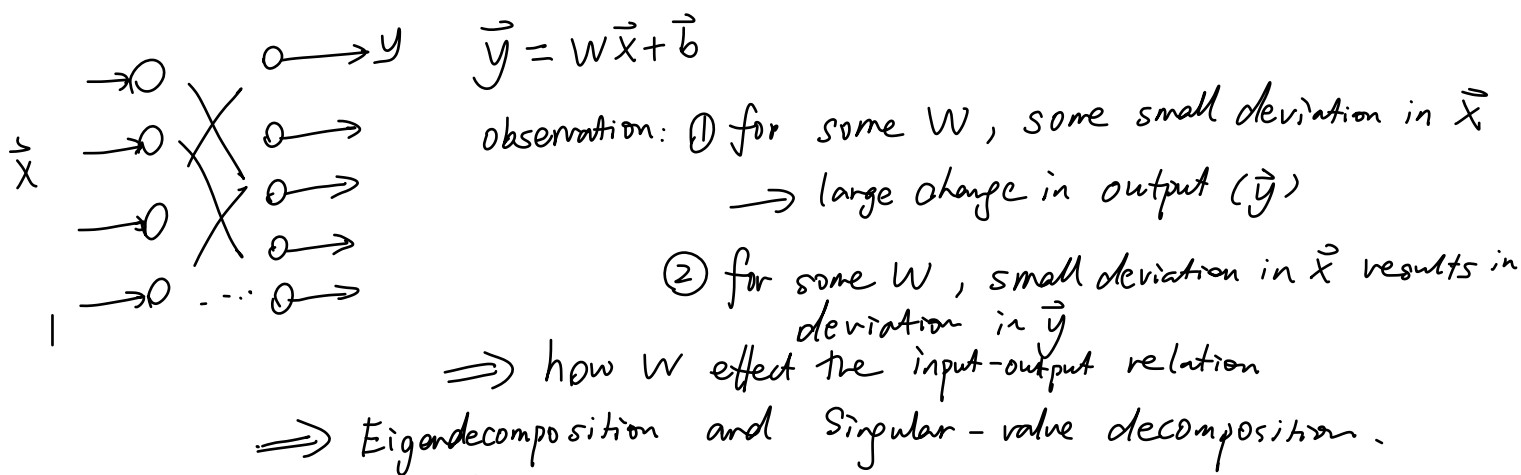


Matrix Decomposition: Eigendecomposition and singular value decomposition.



1. Eigenvalue and eigenvector.

Def: Let $A \in \mathbb{R}^{n \times n}$ be a square matrix, then 1) $\lambda \in \mathbb{R}$ is an eigenvalue of A and $\vec{x} \in \mathbb{R}^n \setminus \{\vec{0}\}$ is the corresponding eigenvector of A , if

$$A\vec{x} = \lambda\vec{x}$$

Remark 1: $\vec{x}$ is a particular direction in space s.t. the linear transformation from $\vec{x}$ to $A\vec{x}$ is equivalent to stretching the vector $\vec{x}$ with parameter λ

Special case: Diagonal matrix (only stretching effect).

$$A_{\text{diag}} = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} \quad A_{\text{diag}}\vec{x} = \begin{bmatrix} \lambda_1 x_1 \\ \vdots \\ \lambda_n x_n \end{bmatrix}, \quad \vec{x} = [x_1, \dots, x_n]^T$$

(larger λ_i , $i \in [1, n]$, larger change in output)

Remark 2: suppose that $\vec{x}$ is an eigenvector of A w.r.t the eigenvalue λ , we have $A(c\vec{x}) = cA\vec{x} = \lambda(c\vec{x})$, for any $c \in \mathbb{R} \setminus \{0\} \Rightarrow c\vec{x}$ is also eigenvector.

Computation of eigenvalue

Theorem: $\lambda \in \mathbb{R}$ is an eigenvalue of $A \in \mathbb{R}^{n \times n}$ iff λ is a root of the characteristic polynomial of A .

$$\downarrow p_A(\lambda)$$

Def. characteristic polynomial.

for $\lambda \in \mathbb{R}$ and a square $A \in \mathbb{R}^{n \times n}$, the characteristic polynomial is defined as.

$$p_A(\lambda) := \det(A - \lambda I)$$

$$= (-1)^n (\lambda - \lambda_1) \dots (\lambda - \lambda_n)$$

$$= \underline{C_0} + \underline{C_1} \lambda + \dots + \underline{C_{n-1}} \lambda^{n-1} + (-1)^n \lambda^n$$

note: $C_0 = \det(A) = \lambda_1 \dots \lambda_n$

$$C_{n-1} = (-1)^{n-1} \cdot \text{tr}(A)$$

$$= (-1)^{n-1} (\lambda_1 + \dots + \lambda_n)$$

trace = sum of eigenvalue

Remark: Formal definition and intuition of det and trace

$\Rightarrow$ Theorem 4.16 / 4.17 and discussion for Figure 4-6.

Special cases

2-D: $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$
 cofactor of a_{11}

3-by-3: $\det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = (-1)^{1+1} a_{11} M_{11} + (-1)^{1+2} a_{12} M_{12} + (-1)^{1+3} a_{13} M_{13}$

$$= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Example

$$A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$$

Eigenvalue

$$\det(A - \lambda I) = \begin{vmatrix} 3-\lambda & 0 \\ 8 & -1-\lambda \end{vmatrix} = 0$$

$$= (3-\lambda)(-1-\lambda) = 0 \quad \lambda_1 = 3 \quad \lambda_2 = -1$$

Eigenvector: $A\vec{x} = \lambda\vec{x}$

① λ_1 : $\begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 3 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \Rightarrow \begin{cases} 3x_1 = 3x_1 \\ 8x_1 - x_2 = 3x_1 \end{cases}$

② λ_2 : $\begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = -1 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \Rightarrow \begin{cases} 3x_1 = -x_1 \Rightarrow x_1 = 0 \\ 8x_1 - x_2 = -x_2 \end{cases} \Rightarrow \begin{cases} 8x_1 = 4x_2 \\ 2x_1 = x_2 \end{cases} \Rightarrow \vec{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$
 x_2 can be anything $\Rightarrow \vec{x} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Equivalent statement

① λ is an eigenvalue of $A \in \mathbb{R}^{n \times n}$: ② $\exists \vec{x} \neq \vec{0}$, s.t. $(A - \lambda I_n)\vec{x} = \vec{0}$

③ $\text{rank}(A - \lambda I_n) < n$; ④ $\det(A - \lambda I_n) = 0$

Several notions about eigenvalue and eigenvector

① Algebraic Multiplicity: Consider $A \in \mathbb{R}^{n \times n}$ with eigenvalue λ_i , the algebraic multiplicity is the number of times the root appears in the characteristic polynomial, denoted by $\text{am}(\lambda_i)$

Example: $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & 2 \\ 2 & 2 & 3 \end{bmatrix}$ $p_A(\lambda) = -(\lambda-1)^2(\lambda-7)$
 repeated eigenvalue.
 $\lambda_1 = 1$ $\lambda_2 = -7$
 $\lambda_3 = 1$

$\text{am}(\lambda=1) = 2$

(2) Eigenspectrum (Spectrum): the set of all eigenvalue

(3) Eigenspace: the set of all eigenvectors of A associated with λ spans the eigenspace of A (denote by E_λ).

associated with $\lambda_1 = 1$, $\vec{x}_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$, $\vec{x}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$, $E_{\lambda_1} = \text{span}(\vec{x}_1, \vec{x}_2)$

$\lambda_2 = 7$, $\vec{x}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \Rightarrow E_{\lambda_2} = \text{span}(\vec{x}_3)$.

(4) Geometric multiplicity: $\text{gm}(\lambda) := \dim(E_\lambda)$

remark: for any eigenvalue λ , $\text{gm}(\lambda) \leq \text{am}(\lambda)$

(5) defective matrix: A square matrix $A \in \mathbb{R}^{n \times n}$ is defective if it possesses fewer than n linear independent eigenvector

Theorem (distinct eigenvalue) The eigenvectors $\vec{x}_1, \dots, \vec{x}_n$ of a matrix $A \in \mathbb{R}^{n \times n}$ with n distinct eigenvalue $\lambda_1, \dots, \lambda_n$ are linearly independent

Corollary: A defective matrix has at least one eigenvalue λ , with $\text{am}(\lambda) > 1$, $\text{am}(\lambda) > \text{gm}(\lambda)$

Theorem 2. If A is symmetric, the eigenvectors to different eigenvalue are orthogonal.

Proof: Consider $\lambda, \mu \in \mathbb{R}$, with $\lambda \neq \mu$, and two vectors $\vec{v}, \vec{w} \in V$ s.t. $A\vec{v} = \lambda\vec{v}$, $A\vec{w} = \mu\vec{w}$, consider dot product: $\langle \cdot, \cdot \rangle$

$\lambda \langle \vec{v}, \vec{w} \rangle = \langle \lambda\vec{v}, \vec{w} \rangle = \langle A\vec{v}, \vec{w} \rangle = (A\vec{v})^T \vec{w} = \vec{v}^T A^T \vec{w} = \langle \vec{v}, A^T \vec{w} \rangle = \langle \vec{v}, A\vec{w} \rangle$
 $= \langle \vec{v}, \mu\vec{w} \rangle = \mu \langle \vec{v}, \vec{w} \rangle$ note that $\lambda \neq \mu \Rightarrow \langle \vec{v}, \vec{w} \rangle = 0$
 $\Rightarrow \vec{v} \perp \vec{w}$

Theorem 3: (Spectral Theorem)

If $A \in \mathbb{R}^{n \times n}$ is symmetric, then there exists an orthonormal basis of the corresponding vector space V , which consists of the eigenvector of A , and each eigenvalue is real.

Remark: for some λ , $\text{am}(\lambda) = \text{gm}(\lambda) > 1 \Rightarrow$ Gram-Schmidt algorithm

Note: consider two eigenvectors $\vec{x}_1, \vec{x}_2$, associated with λ (eigenvalue)

$$A(\alpha \vec{x}_1 + \beta \vec{x}_2) = \alpha \cdot A\vec{x}_1 + \beta A\vec{x}_2 = \alpha \cdot \lambda \vec{x}_1 + \beta \cdot \lambda \vec{x}_2 = \lambda(\alpha \vec{x}_1 + \beta \vec{x}_2)$$

$\Rightarrow$ linear combination of $\vec{x}_1$ & $\vec{x}_2$ is also eigenvector

Theorem 4: Symmetric, positive definite matrices always have positive real eigenvalue.

2. Eigendecomposition.

Let $A \in \mathbb{R}^{n \times n}$, $\lambda_1, \dots, \lambda_n$ being eigenvalue, with the corresponding eigenvector $\vec{p}_1, \dots, \vec{p}_n$, we have

$$A\vec{p}_1 = \lambda_1 \vec{p}_1$$

$$A\vec{p}_2 = \lambda_2 \vec{p}_2$$

$$\vdots$$

$$A\vec{p}_n = \lambda_n \vec{p}_n$$

$$\Rightarrow A \underbrace{[\vec{p}_1 \dots \vec{p}_n]}_P = \underbrace{[\lambda_1 \vec{p}_1 \dots \lambda_n \vec{p}_n]}_{P \cdot D} = \underbrace{[\vec{p}_1 \dots \vec{p}_n]}_P \underbrace{\begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}}_D$$

$$\Rightarrow AP = PD, \text{ if } P \text{ is invertible}$$

$$\underline{A = PDP^{-1}} \quad (A \text{ is symmetric, } A = PDP^T, \text{ Thm 3})$$

Eigendecomposition

Remark: consider $\vec{x} \in \mathbb{R}^n$,

$A \in \mathbb{R}^{n \times n}$ has n linearly independent eigenvectors

$$A\vec{x} = PDP^{-1}\vec{x} \quad \textcircled{1} P^{-1}\vec{x}: \text{transfer } \vec{x} \text{ to the column space of } P^{-1} \text{ (basis change)}$$

$$\textcircled{2} DP^{-1}\vec{x}: \text{stretching } P^{-1}\vec{x} \text{ with } D \text{ (eigenvalue)}$$

$$\textcircled{3} PDP^{-1}\vec{x}: \text{transfer the stretched vector } DP^{-1}\vec{x} \text{ to column space of } P$$

$$\text{In particular, } A^k = (PDP^{-1})^k = \underbrace{(PDP^{-1})(PDP^{-1}) \dots (PDP^{-1})}_I = PD^kP^{-1}$$

$\rightarrow$ Note: "Similar" to pure stretching the vectors, but under different basis
From the perspective of similarity:

A matrix $A \in \mathbb{R}^{n \times n}$ is diagonalizable if it is similar to a diagonal matrix $\Rightarrow \exists D \in \mathbb{R}^{n \times n}, D = P^{-1}AP \leftarrow A = PDP^{-1}$

$\Rightarrow$ Cholesky Decomposition (Symmetric, positive definite matrix)

Sec. 4.3

3. Singular Value decomposition.

$$A \in \mathbb{R}^{m \times n} \quad \text{rank}(A) = r \leq \min(m, n)$$

$$\begin{matrix} n \\ m \end{matrix} \begin{matrix} \boxed{A} \end{matrix} = \begin{matrix} m \\ m \end{matrix} \begin{matrix} \boxed{U} \end{matrix} \begin{matrix} n \\ m \end{matrix} \begin{matrix} \boxed{\Sigma} \end{matrix} \begin{matrix} n \\ n \end{matrix} \begin{matrix} \boxed{V^T} \end{matrix}$$

any matrix!

• $U \in \mathbb{R}^{m \times m}$ with orthogonal column vectors
 $\vec{u}_i, i=1, \dots, m$

$V \in \mathbb{R}^{n \times n}$ $\vec{v}_j, j=1, \dots, n$.

$\Sigma \in \mathbb{R}^{m \times n}$, with $\Sigma_{ii} = \sigma_i \geq 0$, $\Sigma_{i,j} = 0$, with $i \neq j$.

• $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r \geq 0 \leftarrow$ singular value.

$\vec{u}_i$: left-singular vectors $\vec{v}_j$: right-singular vectors

Relation with eigendecomposition.

For symmetric, positive definite matrix S

Eigendecomposition: $S = P D P^T = \underline{P} \underline{D} \underline{P}^T$ (P can be orthonormal).

SVD : $S = \underline{U} \underline{\Sigma} \underline{V}^T$ $U = P = V$, $D = \Sigma$

For symmetric positive definite matrix, SVD and eigendecomposition can be the same, i.e. decompose the matrix (linear transformations) as rotation - scale - rotation.

Construction of SVD (U, Σ, V)

✓ ① construction of V

$$A^T A = (\underline{U} \underline{\Sigma} \underline{V}^T)^T (\underline{U} \underline{\Sigma} \underline{V}^T) = \overset{\substack{\uparrow \\ \text{orthonormal.}}}{\underline{V} \underline{\Sigma}^T \underline{U}^T \underline{U} \underline{\Sigma} \underline{V}^T} = \underline{V} \underline{\Sigma}^T \underline{\Sigma} \underline{V}^T$$

$$= \underline{V} \begin{bmatrix} \sigma_1^2 & & \\ & \ddots & \\ & & \sigma_n^2 \end{bmatrix} \underline{V}^T$$

Eigendecomposition of $A^T A$

$\Rightarrow \underline{V}^T = \underline{P}^T$ (eigenvector of $A^T A$), $\underline{\sigma}_i^2 = \underline{\lambda}_i$ (eigenvalue of $A^T A$)

✓ ② construction of U .

$$A A^T = \underline{U} \underline{\Sigma} \underline{V}^T (\underline{U} \underline{\Sigma} \underline{V}^T)^T = \underline{U} \underline{\Sigma} \underbrace{\underline{V}^T \underline{V}}_{\underline{I}} \underline{\Sigma}^T \underline{U}^T = \underline{U} \underline{\Sigma} \underline{\Sigma}^T \underline{U}^T = \underline{U} \begin{bmatrix} \sigma_1^2 & & \\ & \ddots & \\ & & \sigma_m^2 \end{bmatrix} \underline{U}^T$$

$\Rightarrow AA^T$ is symmetric $\Rightarrow U$ contains (orthonormal) eigenvector of AA^T
 $\sigma_i^2 = \lambda_i$ (eigenvalue of AA^T)

AA^T and $A^T A$ have the same eigenvalues (not eigenvectors)

Proof: Consider $A \in \mathbb{R}^{m \times n}$, $\lambda \neq 0$ and $\vec{v}$ being eigenvector of $A^T A$

$$A^T A \vec{v} = \lambda \vec{v}, \text{ left multiplication of } A$$

$$\star AA^T A \vec{v} = A(A^T A \vec{v}) = A(\lambda \vec{v}) = \lambda A \vec{v}$$

$\Rightarrow \lambda$ is the eigenvalue of AA^T with the corresponding eigenvector

$A \vec{v}$
 $\Rightarrow$ any non-zero eigenvalue λ of $A^T A$ is also an eigenvalue of that of AA^T

Similarly, consider $\beta \neq 0$ being eigenvalue of AA^T , with eigenvector $\vec{u}$, s.t. $AA^T \vec{u} = \beta \vec{u}$, multiplying A^T on both side.

$$A^T(AA^T \vec{u}) = A^T A \vec{u} = \beta A^T \vec{u} \Rightarrow \beta \text{ is an eigenvalue of } A^T A$$

$\Rightarrow \sigma_1^2, \dots, \sigma_n^2$ and $\sigma_1^2, \dots, \sigma_m^2$ are consistent

Computational method: ① $A = U \Sigma V^T \Rightarrow \underline{A} \underline{V} = \underline{U} \underline{\Sigma} \underline{V}^T \underline{V} = \underline{U} \underline{\Sigma}$

$$\Rightarrow \star A \cdot [\vec{v}_1 \dots \vec{v}_n] = [\vec{u}_1 \dots \vec{u}_n] \begin{bmatrix} \sigma_1 & & 0 \\ & \ddots & \\ 0 & \dots & 0 \end{bmatrix}$$

$m \times n$ $n \times n$ $n \times n$

$$A \vec{v}_i = \sigma_i \vec{u}_i, \quad i=1, \dots, r \Rightarrow \vec{u}_i = \frac{1}{\sigma_i} A \vec{v}_i$$

remark: for $i = r+1, \dots, n$, $\vec{u}_i$ can be any unit vector which is perpendicular to $\vec{u}_i$ with $i=1, \dots, r$

Exercise: $A = \begin{bmatrix} 2 & 1 \end{bmatrix}$, perform SVD

① compute orthonormal eigenvector of $A^T A$.

$$A^T A = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$$

$$\det(A^T A - \lambda I) = \begin{vmatrix} 4-\lambda & 2 \\ 2 & 1-\lambda \end{vmatrix} = (4-\lambda)(1-\lambda) - 4$$

$$= \lambda^2 - 5\lambda + 4 - 4 = \lambda^2 - 5\lambda = \lambda(\lambda - 5)$$

$$\lambda_1 = 5, \quad \lambda_2 = 0$$

② $\|A - \hat{A}(k)\|_2 = \sigma_{k+1}$ (how large is the approximation error).

sec 4.2 - 4.6